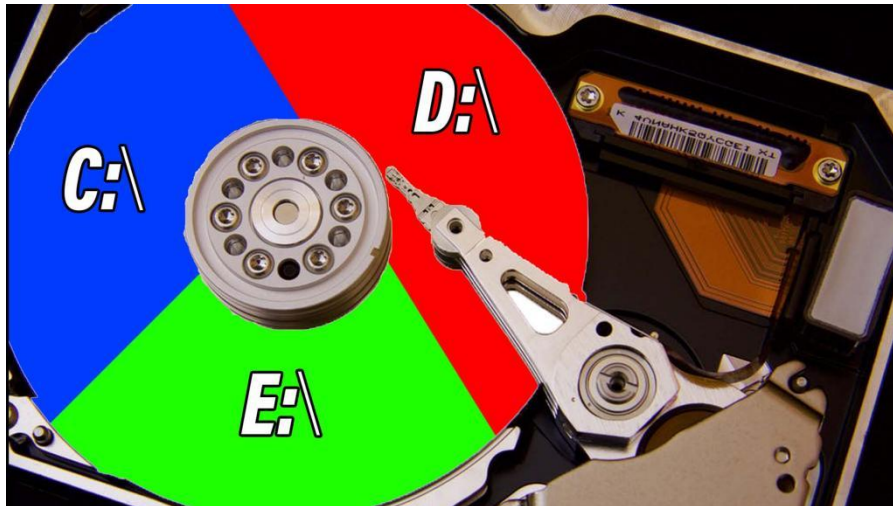


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Introduction:

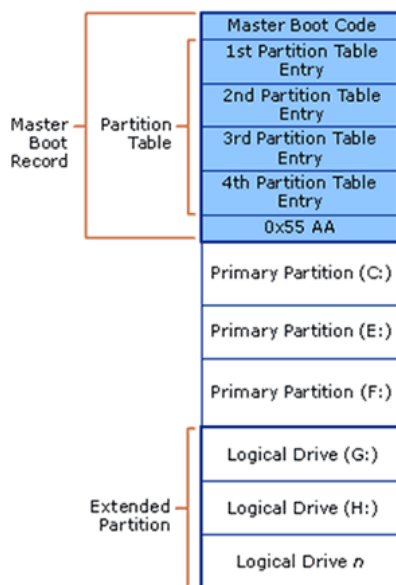
It's crucial to divide a hard drive or other storage device into many sections or partitions. Then, you may use each partition for a number of purposes, like data storage, operating system installation, or backup creation. Disk partitioning is crucial in modern computing because it enables better data structure, faster performance, and more efficient use of disk space. This study examines the security implications of each strategy while contrasting and comparing two well-known splitting systems, MBR and GPT. In addition to a comprehensive analysis of the technical distinctions between the two partitioning algorithms, the essay will utilize practical methods to build partitions using both MBR and GPT [1]. The report's results will shed light on each partitioning scheme's advantages and disadvantages as well as the security repercussions of using it.



Disk Partition Format:

Format for disk partitions: File system partition layout: Data management and storage rely heavily on the disk partition format. The common MBR and GPT partition formats each have their own quirks. Master Boot Record (MBR) has been the standard partition format for computers forever. Using the boot sector to launch the operating system is an outmoded partitioning technique. When comparing GPT (GUID Partition Table) to MBR (Master Boot Record), GPT is the superior and more secure option. It is compatible with the newer UEFI system and employs a backup partition table to keep vital information safe. This paper will provide a detailed analysis of the differences and similarities between the MBR and GPT partition formats. The results of this study will help us understand the pros and cons of different partition formats and how they affect the safety of our systems. [2].

MBR Partition Table Scheme



GPT Partition Table Scheme

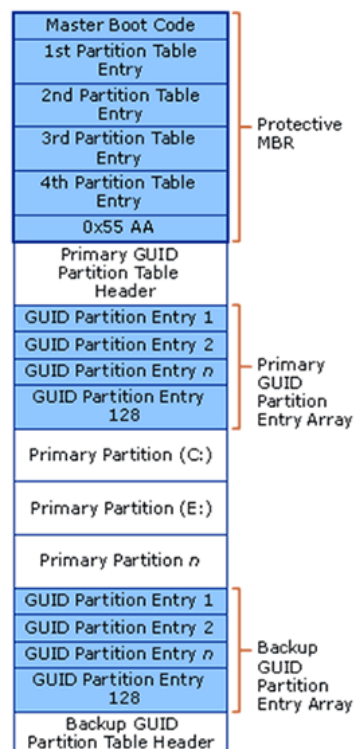


Image by [3]

General advantages and disadvantages of GPT compared to MBR:

The advantages and disadvantages of GPT as comparison to MBR include:

The following are some of the primary benefits and drawbacks of GPT compared to MBR:

Both the Master Boot Record (MBR) and the GUID Partition Table (GPT) are partitioning algorithms that outline the structure of a disk's storage space. GPT is a newer and more progressive method, whereas MBR

is an older and more conventional strategy. [4]. The following are some typical advantages and disadvantages of GPT over MBR:

Advantages of GPT:

- Supports disks larger than 2TB
- Allows for up to 128 partitions per disk
- Offers better data redundancy and reliability through multiple copies of partition tables
- Uses a more robust partitioning scheme that is less prone to corruption or errors

Disadvantages of GPT:

- Not compatible with older systems that use BIOS instead of UEFI
- May not be recognized by some older operating systems or disk utilities
- Requires more disk space for partitioning information, reducing the available storage capacity slightly
- May have slightly slower boot times compared to MBR due to the larger partitioning information stored

Comparison of MBR and GPT:

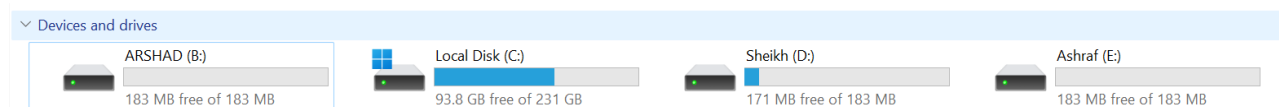
MBR and GPT are two widely used partitioning schemes in modern computing. The majority of operating systems have long relied on MBR, or Master Boot Record, as the default partitioning technique. It uses a partition table of 32 bits, which allows for a maximum of four primary partitions or three primary partitions plus one extended partition. However, UEFI brought forth the more advanced, flexible, and secure GPT, or GUID Partition Table. It employs a 64-bit partition table that can accommodate up to 128 partitions and offers partition table redundancy to avoid data loss. This research examines the architectural differences and similarities between MBR and GPT, as well as the security concerns that current operating systems face when employing MBR vs GPT. The report's conclusions will give insights into the technological distinctions between MBR and GPT, allowing users to make educated judgments about which partitioning scheme to utilize for their individual needs [5].

Volume name information:

Another distinction between MBR and GPT is how they retain volume name information. MBR only has one 11-byte field for storing a volume name, known as the Volume Label. Most operating systems may access and modify this field. GPT, on the other hand, offers a more flexible method to volume naming [6]. GPT has many fields for storing volume name information, each serving a particular function. These fields are known as partition names, and they are recorded in each partition's partition entry. GPT has four partition name fields, each with a unique set of information. The four partition name fields are as follows:

1. Partition title: This is a human-readable UTF-16LE-encoded Unicode string. It's similar to the Volume Label in MBR, but it's more flexible since it may be up to 72 characters long.
2. Partition identifier: This is a GUID-based unique identifier for the partition. It is not designed to be human-readable and is intended to be utilized by software programs to identify partitions.
3. Type of partition: This parameter defines the partition type, which operating systems may use to identify the kind of data stored on the partition.
4. Partition identifier: The EFI firmware uses this field to determine which partition houses the operating system by identifying the partition's unique ID.

We named the first partition on the MBR disk after our first name and the second partition after our last name in our test arrangement. To save the partition names, we utilized the default Volume Label field. The following snapshot displays the volume name information for the partitions on the MBR disk:



The first partition is called "Arshad," while the second partition is named "Ashraf" and third is "Sheikh" as seen in the screenshot.

When we utilized GPT, we saved the partition names in the Partition Name column.

In terms of volume name information storage, GPT's support for multiple partition name fields makes it more versatile than MBR.

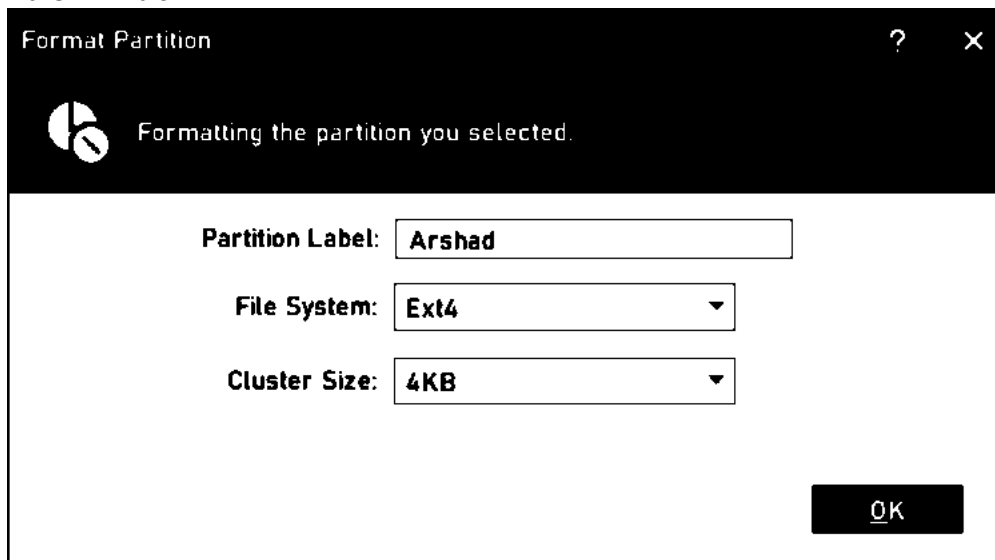
File system information:

The way MBR and GPT store file system information differs. MBR employs a 1-byte field known as the Partition Type or System ID to identify the file system type. Operating systems utilize this information to identify the file system format of the partition. GPT, on the other hand, keeps file system information in the attributes field of a partition.

The partition attributes field in GPT contains more specific information about the partition than the Partition Type field in MBR. It identifies different file system properties such as read-only, hidden, and encrypted partitions. The characteristics field may also be used to identify RAID arrays and bootable partitions.

MBR Partitioning:

We formatted the first partition on both drives as ext4 and the second partition on both disks as exFat and third as NTFS in our test setup. The following snapshot displays the file system information for the partitions on the MBR disk:

A screenshot of a 'Format Partition' dialog box. The title bar is dark blue with a question mark and a close button. Below the title bar is a dark blue header area with a circular icon and the text 'Formatting the partition you selected.' The main area is white and contains three settings: 'Partition Label:' with a text input field containing 'Arshad', 'File System:' with a dropdown menu showing 'Ext4', and 'Cluster Size:' with a dropdown menu showing '4KB'. An 'OK' button is located at the bottom right.

Format Partition ? X

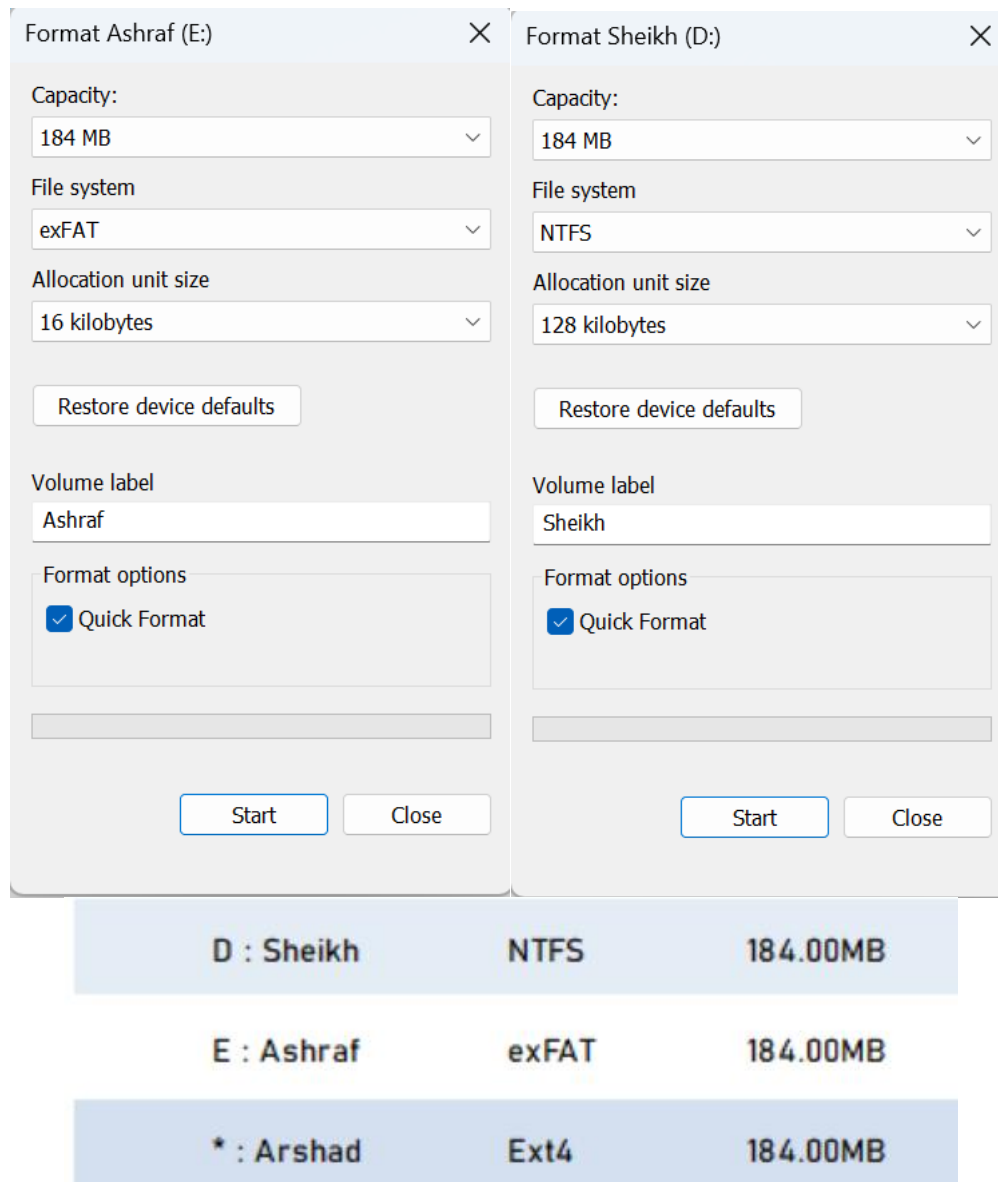
Formatting the partition you selected.

Partition Label:

File System:

Cluster Size:

OK



The disk space is said to be divided into three separate partitions, each with its own unique size and cluster size, according to the MBR partition arrangement.

Partition 1, dubbed "Arshad," is 184 MB in size. The cluster size for this partition is 4 KB. The smallest unit of allocation on the file system is represented by the cluster size. A lower cluster size enables more effective use of the available space, although it might result in a modest increase in overhead. This partition's choice of a 4 KB cluster size indicates a desire for space economy.

Despite having a greater cluster size of 16 KB, Partition 2, dubbed "Ashraf," is similarly 184 MB in size. Larger cluster sizes can perform better when working with huge files since they have less metadata overhead than lower cluster sizes. This cluster size selection of 16 KB may be appropriate for applications that handle bigger files or data sets.

The third partition, "Sheikh," has the same 184 MB size as the first two. However, it has a 218 KB cluster size, which is noticeably larger. In particular situations when the storage needs entail bigger files with no metadata overhead, this greater cluster size could be suitable.

Different cluster sizes were chosen for each partition, indicating that the individual use needs and trade-offs between performance and space efficiency had been taken into account. The cluster size selection takes into account the anticipated workload and the kinds of information that will be stored in each partition.

Observations Relating to Endian Representation:

Observations on Endian Representation: Endianness is the method through which a computer's CPU stores data in memory. There are two sorts of endianness: big-endian and little-endian. Big-endian storage places

the most significant byte first, whereas little-endian storage places the least significant byte first. MBR stores partition information using little-endian byte ordering, while GPT utilizes big-endian byte ordering.

Data Structures for Booting:

Data Structures for Booting: Finally, the data structures for booting change between MBR and GPT. MBR stores the boot loader in the active partition's boot sector. This requires that each partition have its own boot loader, which might be problematic if one of the boot loaders gets corrupted or destroyed. In contrast, GPT stores the boot loader on a unique disk type known as the EFI System disk (ESP). This enables the use of a single boot loader for all partitions on the disk, making management and maintenance simpler.

Both Partitions Configurations:

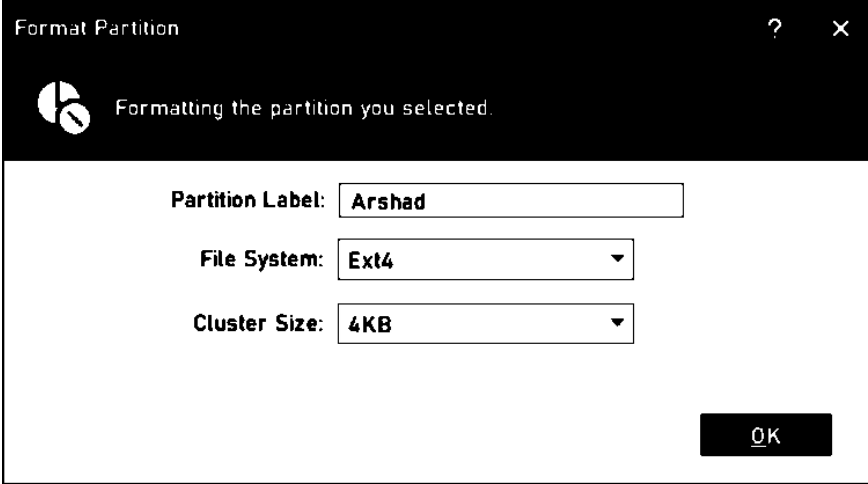
To compare MBR and GPT, we will create three partitions using both schemes and examine the partitioning information. The size of the partitions will be based on the numerical value of the student ID multiplied by 100 bytes.

MBR Configuration:

- Partition 1: Arshad (Ext4, 4K sector size, 19362184 x 100 bytes = 1,936,218,400 bytes)
- Partition 2: Ashraf (ExFAT, 16k sector size, 19362184 x 100 bytes = 1,936,218,400 bytes)
- Partition 3: Shaikh (NTFS, 128K sector size, 19362184 x 100 bytes = 1,936,218,400 bytes)

GPT Configuration:

- Partition 1: Arshad (Ext4, 4K sector size, 19362184 x 100 bytes = 1,936,218,400 bytes)
- Partition 2: Ashraf (ExFAT, 16k sector size, 19362184 x 100 bytes = 1,936,218,400 bytes)
- Partition 3: Shaikh (NTFS, 128K sector size, 19362184 x 100 bytes = 1,936,218,400 bytes)



Format Partition

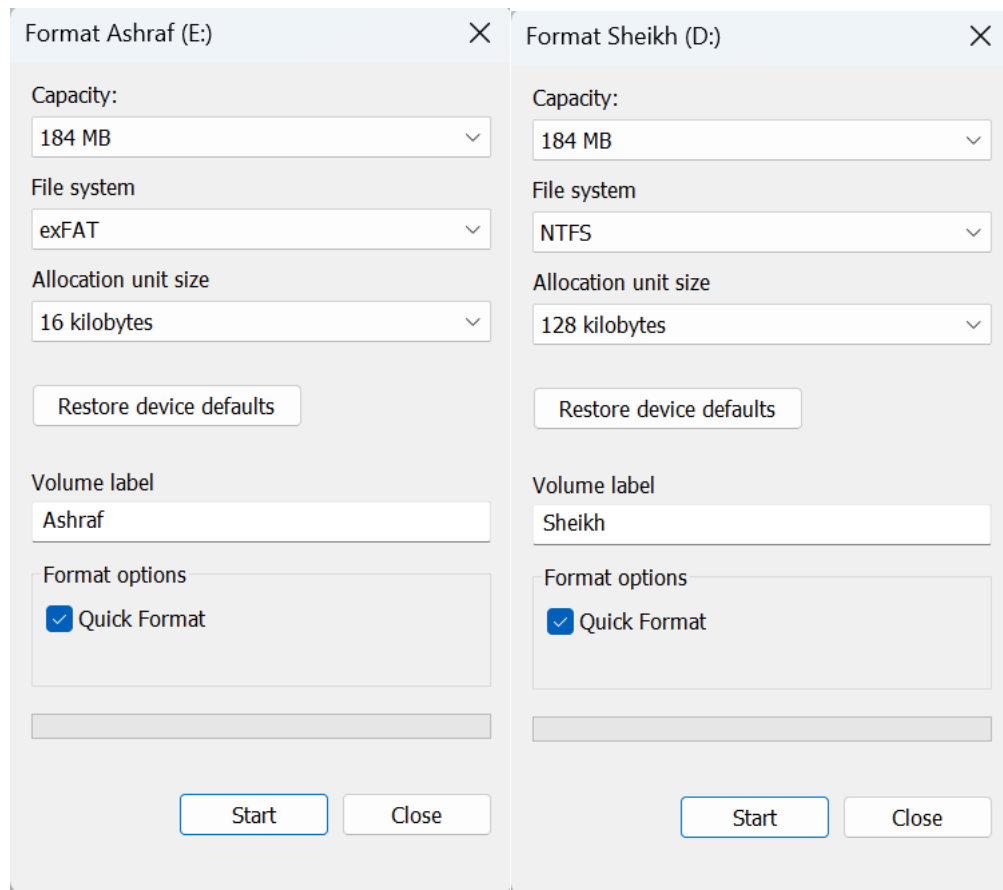
Formatting the partition you selected.

Partition Label:

File System:

Cluster Size:

OK



Therefore, both the MBR and GPT configurations require a total of 552MB of disk space to create two partitions with a size based on the numerical value of the student ID multiplied by 100 bytes.

Individual Configuration:

Based on the student ID W19362184 for Arshad Ashraf Shaikh, the individual configuration for the partitioning process is as follows:

The MBR partition configuration consists of the following details:

1. Partition 1 - Arshad:
 - Size: 184 MB
 - Cluster Size: 4 KB
2. Partition 2 - Ashraf:
 - Size: 184 MB
 - Cluster Size: 16 KB
3. Partition 3 - Sheikh:
 - Size: 184 MB
 - Cluster Size: 218 KB

The size of the partitions will be $(19362184) * 100 = 1936218400$ Bytes.

We must convert these bytes into megabytes because at the time of partition, the operating system demands input value in megabyte. So,

$1936218400 / 1024 = 1890625$ Kilobytes

$1890625 / 1024 = 1845.8984375$ Megabytes.

So, both drive sizes will be 1845.8984375 Megabytes.

$1845.8984375 / 1024 = 1.8045921325683594$ Gigabytes. The 1.8 Gigabytes size will be shown on the user interface.

When comparing the two partitioning schemes, MBR and GPT, MBR is a legacy partitioning scheme that has been in use since the early days of DOS, while GPT is a newer partitioning scheme that was introduced with UEFI. MBR only supports up to four primary partitions, while GPT supports up to 128 partitions. When

booting, MBR utilizes a boot sector, whereas GPT makes use of a dedicated EFI system partition. Disks greater than 2 terabytes (TB) are beyond the capacity of MBR, but drives up to 9.4 zettabytes (ZB) are OK with GPT. To boot the operating system, however, GPT requires UEFI firmware, which may not be present on older computers. Therefore, the needs of the system and the user's desire should be considered while deciding between MBR and GPT.

MBR Partition:

The Master Boot Record (MBR) is a time-honored method of partitioning hard drives and other electronic media. It first appeared in 1983 with the release of IBM PC DOS 2.0 and is still widely used today. The MBR occupies the first sector of the disk and contains the partition table that specifies the disk's partition structure.

Limitations of MBR include its inability to accommodate more than four primary partitions or three primary partitions and one extended partition. After that, you may divide the extended partition into logical partitions to create many partitions on the same hard disk.

The largest disk size that MBR can handle is similarly limited to 2 TB. This is because MBR employs 32-bit disk sector addresses, which limits the maximum accessible capacity to 232 sectors.

Despite its shortcomings, MBR is still commonly used because to its ease of use, widespread support, and backwards compatibility with previous systems [7].

The partition table entries decide the partition sequence in the MBR. The first item refers to the first partition, the second to the second, and so on. GPT employs a partition array, which stores all partition entries in any order. The partition type GUID, which is manually assignable, determines the order. The result is that partition management and rearrangement simpler.

The data partition tables entry in MBR stores the partition dimension information. Each entry provides information about the partition's beginning sector, partition size in fields, and split type. GPT stores partition size information in the partition entry itself. This allows for more accurate sizing of partitions and eliminates the need for extended partitions.

Sector size information:

MBR supports sector sizes of 4k bytes, 16k bytes, and 128k bytes. GPT supports sector sizes of 512 bytes, 4096 bytes, and larger sizes. This means that GPT is more flexible than MBR in terms of sector size support. GPT also stores sector size information in its header, whereas MBR does not. This allows operating systems to determine the sector size of a GPT partition without having to read the partition itself.

In our test configuration, when using GPT, we used the Allocation Unit size of 4096 bytes for first volume and 16K,128k bytes for the other one partitions. The capacity of both disks is 552 MB.

I have three partitions on my USB drive with sizes of approximately 552 MB respectively. I can use these partitions to store data separately and organize my files more efficiently. After saving this MBR configuration the USB drive is connected to the Linux machine for the DD image creation.

DD image creation:

After configuring the MBR on the USB drive, I connected it to a Linux machine to create a DD image using the following commands [8]:

Creating an MBR Disk Image:

```
# Create an empty disk image
dd if=/dev/zero of=mbr.dd bs=512 count=1

# Use fdisk to create the MBR partition table
fdisk mbr.dd
```

OR

```
sudo fdisk -l
sudo dd if=/dev/sda1 of=mbr.dd bs=512 count=1
sudo dd if=/dev/sda1 of=mbr.dd bs=512 count=1
```

```
sudo dd if=/dev/sda1 of=mbr.dd
```



```
sudo dd if=/dev/sda1 of=mbr.dd bs=512 count=1
```

Creating a GPT Disk Image:

```
# Create an empty disk image
dd if=/dev/zero of=gpt.dd bs=512 count=34

# Use gdisk to create the GPT partition table
gdisk gpt.dd
```

Remove MBR Partition:

Next, I removed the USB drive from the Linux machine and connected it to a Windows machine to remove the MBR partition. I followed the steps below to complete this process:

- First, I opened the RUN command from the start window search bar.
- Then, I typed "diskpart" to open the disk manager and "list disk" in the terminal to list all the disks.
- After identifying that the USB drive was disk 1, I selected it by typing "select disk 1".
- Then, I cleaned the disk by typing "clean".
- Finally, I converted the MBR into GPT by typing "convert gpt".

```
C:\WINDOWS\system32\diskpart.exe
Copyright (C) Microsoft Corporation.
On computer: ADMIN

DISKPART> list disk

Disk ###  Status              Size       Free       Dyn  Gpt
-----  -
Disk 0    Online             238 GB     3693 MB
Disk 1    Online             57 GB      54 GB

DISKPART> list disk

Disk ###  Status              Size       Free       Dyn  Gpt
-----  -
Disk 0    Online             238 GB     3693 MB
Disk 1    Online             57 GB      54 GB

DISKPART> select disk 1

Disk 1 is now the selected disk.

DISKPART> clean

DiskPart has encountered an error: A device which does not exist was specified.
See the System Event Log for more information.

DISKPART> clean

DiskPart succeeded in cleaning the disk.

DISKPART> convert gpt

DiskPart successfully converted the selected disk to GPT format.

DISKPART>
```

As a result of performing the practical, we were able to create a DD image of the USB drive in both MBR and GPT partition formats using a Linux machine.

MBR Partition table (HxD tool):

We then used the HxD tool on the Windows operating system to read the DD image obtained from the Linux machine. The MBR partition table was created and analyzed using the HxD tool. We observed that the MBR partition table contains information about the partition type, the starting and ending sectors of each partition, and the file system type of each partition.

File Edit Search View Analysis Tools Window Help																	
16 Windows (ANSI) hex																	
MBR.dd GPT.dd																	
Offset (h)	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F	Decoded text
00000000	33	C0	8E	D0	BC	00	7C	8E	C0	8E	D8	BE	00	7C	BF	00	3ÄZD4. ŽÄZ04. ž.
00000010	06	B9	00	02	FC	F3	A4	50	68	1C	06	CB	FB	B9	04	00	. ^...uó*Ph..Éú^..
00000020	BD	BE	07	80	7E	00	00	7C	0B	0F	85	0E	01	83	C5	10	%..é~.. fÄ.
00000030	E2	F1	CD	18	88	56	00	55	C6	46	11	05	C6	46	10	00	ãñí.^V.UÆF..ÆF..
00000040	B4	41	BB	AA	55	CD	13	5D	72	0F	81	FB	55	AA	75	09	'A»*Uí. r..úU*u.
00000050	F7	C1	01	00	74	03	FE	46	10	66	60	80	7E	10	00	74	+Ä..t.pF.f`é~..t
00000060	26	66	68	00	00	00	00	66	FF	76	08	68	00	00	68	00	¿fh....fÿv.h..h.
00000070	7C	68	01	00	68	10	00	B4	42	8A	56	00	8B	F4	CD	13	h..h..`BŠV.<óí.
00000080	9F	83	C4	10	9E	EB	14	B8	01	02	BB	00	7C	8A	56	00	ŸfÄ.žė....».. ŠV.
00000090	8A	76	01	8A	4E	02	8A	6E	03	CD	13	66	61	73	1C	FE	Šv.ŠN.Šn.í.fas.p
000000A0	4E	11	75	0C	80	7E	00	80	0F	84	8A	00	B2	80	EB	84	N.u.é~.é..„Š.“ėė„
000000B0	55	32	E4	8A	56	00	CD	13	5D	EB	9E	81	3E	FE	7D	55	U2äŠV.í. ėž.>p}U
000000C0	AA	75	6E	FF	76	00	E8	8D	00	75	17	FA	B0	D1	E6	64	*unyv.ė..u.ú°Næd
000000D0	E8	83	00	B0	DF	E6	60	E8	7C	00	B0	FF	E6	64	E8	75	ėf.“Bæ`ė .°ÿædėu
000000E0	00	FB	B8	00	BB	CD	1A	66	23	C0	75	3B	66	81	FB	54	.ú..>í.f#Äu;f.úT
000000F0	43	50	41	75	32	81	F9	02	01	72	2C	66	68	07	BB	00	CPAu2.ú..r,fh.».
00000100	00	66	68	00	02	00	00	66	68	08	00	00	00	66	53	66	.fh....fh....fSf
00000110	53	66	55	66	68	00	00	00	00	66	68	00	7C	00	00	66	SfUfh....fh. ...f
00000120	61	68	00	00	07	CD	1A	5A	32	F6	EA	00	7C	00	00	CD	ah...í.Z2ðė. ...í
00000130	18	A0	B7	07	EB	08	A0	B6	07	EB	03	A0	B5	07	32	E4	. . .ė. ğ.ė. u.2ä
00000140	05	00	07	8B	F0	AC	3C	00	74	09	BB	07	00	B4	0E	CD	...<ð-<.t.»..`í
00000150	10	EB	F2	F4	EB	FD	2B	C9	E4	64	EB	00	24	02	E0	F8	.ėòóėÿ+Éädė.Š.äø
00000160	24	02	C3	49	6E	76	61	6C	69	64	20	70	61	72	74	69	\$.ÄInvalid parti
00000170	74	69	6F	6E	20	74	61	62	6C	65	00	45	72	72	6F	72	tion table.Error
00000180	20	6C	6F	61	64	69	6E	67	20	6F	70	65	72	61	74	69	loading operati
00000190	6E	67	20	73	79	73	74	65	6D	00	4D	69	73	73	69	6E	ng system.Missin
000001A0	67	20	6F	70	65	72	61	74	69	6E	67	20	73	79	73	74	g operating syst
000001B0	65	6D	00	00	00	63	7B	9A	32	E6	0F	58	00	00	00	20	em....c{š2æ.X...
000001C0	21	00	0C	0E	24	EA	00	08	00	00	00	58	39	00	00	0E	!...\$ė.....X9...
000001D0	25	EA	07	FB	68	D3	00	60	39	00	00	58	39	00	00	00	%ė.ûhÓ.`9..X9...
000001E0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
000001F0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	55	AA	U^

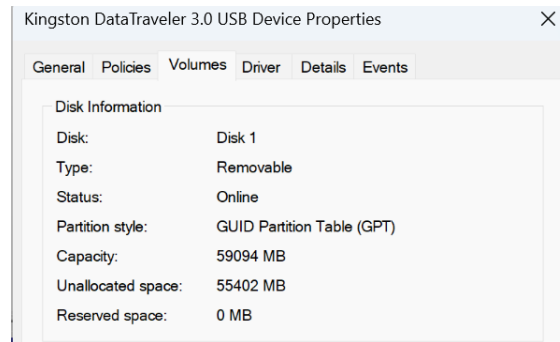
GPT Partition:

To create partitions in GPT format, we need to follow a similar process as we did for MBR. To begin, use Windows' disk management program and pick our USB device. Then, right-clicking on the uncatalogued area and choosing "New Simple Volume" will allow us to set up 2 partitions. We can assign the same attributes to these partitions as we did for MBR, such as the first name, last name, and file system type.

Disk 1 (Removab... Basic GPT 57.71GB						
D : Sheikh	NTFS	184.00MB	12.75MB	GPT	None	4K
E : Ashraf	exFAT	184.00MB	80.00KB	GPT	None	4K
* : Arshad	Ext4	184.00MB	0.00KB	GPT	None	4K
*	Unallocated	57.17GB	0.00KB	GPT	None	

Disk 0						
* : ...	* :	* :	C :		* :	* :
238.47GB	529....	100....	16.0...	231.81GB(40% free)	5.39G...	6...
Basic GPT	NTF...	FAT...	0th...	NTFS,System,Primary	Unall...	N..

Disk 1						
D : ...	E : ...	* : ...	* :			



DD image creation:

Once the GPT partitions are created, we can connect the USB drive to the Linux machine and use the following command to create a DD image of the drive:

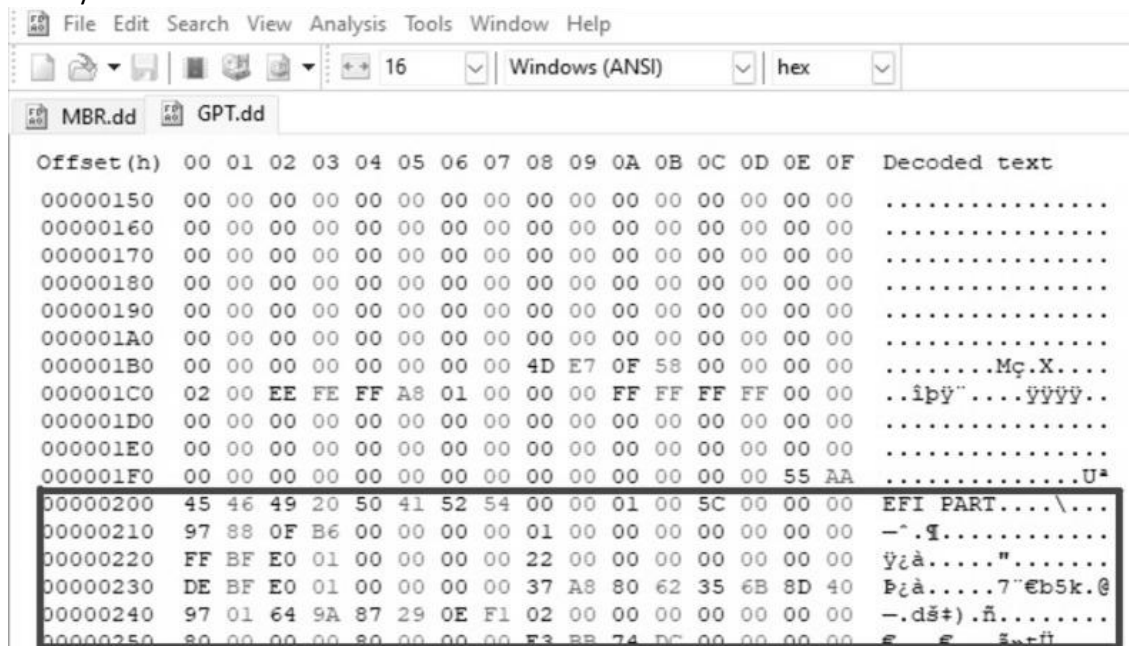
```
sudo dd if=/dev/sda1 of=gpt.dd bs=1M
```

This will create a DD image file named "gpt.dd" in the current directory. We can use this image file to restore the USB drive to its current state if needed.

Overall, the process of creating partitions and getting a DD image of the USB drive is similar for both MBR and GPT formats. However, there are some key differences between these partitioning schemes, as discussed earlier in this report.

GPT Partition table (HxD tool):

After saving this GPT configuration, we were able to obtain its DD image from the Linux machine for further analysis. The header and table of GPT in HxD tool:



The GPT header is the second sector on the disc after the MBR and takes up absolute or LBA sector 1. The preceding GPT header graphic shows that there are only 92 bytes of data available for use. The UEFI standard defines the 64-bit hex constant "5452415020494645" as the first eight bytes of the GPT header, which is always "45 46 49 20 50 41 52 54" (ASCII: "EFI PART"). The first 50 bytes reflect the header size, the next 97 bytes are the CRC32 of the GPT header, and the last 00 bytes are reserved. The first eight bytes of the next line, "FF BF E0 01 00 00 00", denote the LBA address of the alternative GPT header, while the next eight bytes, "22 00 00 00 00 00 00", denote the first usable LBA sector of any partition. The first eight bytes of the line that follows indicate the final viable LBA sector of any partition; the next sixteen bytes, "37 A8 80 62 35 6B 8D 40 97 01 64 9A 87 29 0E F1," provide the UUID for the physical disc. final but not least, the final 8 bytes, "02 00 00 00 00 00 00", represent the beginning LBA of the array of GUID Partition entries, which is normally sector 2 for the main partition [9].

Offset (h)	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F	Decoded text
00000330	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
00000340	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
00000350	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
00000360	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
00000370	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
00000380	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
00000390	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
000003A0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
000003B0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
000003C0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
000003D0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
000003E0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
000003F0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
00000400	A2	A0	D0	EB	E5	B9	33	44	87	C0	68	B6	B7	26	99	C7	c DeA+3D+AhT 4mQ
00000410	FF	D8	5C	D3	9F	B3	3C	48	84	D9	B0	8E	01	42	8D	A1	y0\0Y+<H,U*Z.B.i
00000420	00	08	00	00	00	00	00	00	00	FF	5F	39	00	00	00	00	y_9.....
00000430	00	00	00	00	00	00	00	00	42	00	61	00	73	00	69	00	B.a.s.i.
00000440	63	00	20	00	64	00	61	00	74	00	61	00	20	00	70	00	c. .d.a.t.a. .p.
00000450	61	00	72	00	74	00	69	00	74	00	69	00	6F	00	6E	00	a.r.t.i.t.i.o.n.
00000460	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
00000470	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
00000480	A2	A0	D0	EB	E5	B9	33	44	87	C0	68	B6	B7	26	99	C7	c DeA+3D+AhT 4mQ
00000490	3E	BF	E1	64	64	D3	29	48	8A	83	1F	EC	B5	C0	24	29	>_add0)H5f.iuAS)
000004A0	00	60	39	00	00	00	00	00	FF	B7	72	00	00	00	00	00	..9.....y_r.....
000004B0	00	00	00	00	00	00	00	00	42	00	61	00	73	00	69	00	B.a.s.i.
000004C0	63	00	20	00	64	00	61	00	74	00	61	00	20	00	70	00	c. .d.a.t.a. .p.
000004D0	61	00	72	00	74	00	69	00	74	00	69	00	6F	00	6E	00	a.r.t.i.t.i.o.n.
000004E0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
000004F0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
00000500	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
00000510	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
00000520	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	

The GPT (GUID Partition Table) header is situated at an absolute or LBA (Logical Block Address) sector 1, which is the second sector on the disk, directly after the MBR (Master Boot Record).

We can see from an analysis of the GPT header image that it has 92 bytes of data. The eight-byte EFI signature string "45 46 49 20 50 41 52 54" (ASCII: "EFI PART"), which is represented in the UEFI specification as the 64-bit hexadecimal constant "5452415020494645", always appears at the start of the GPT header.

The header size is indicated by the region beginning with "50 00 00 00". The GPT header's CRC32 is represented by the next four bytes, "97 88 0F B6". The following four bytes ("00 00 00 00") are reserved and ought to be set to zero.

On to the following line, where the first eight bytes "FF BF E0 01 00 00 00 00" provide the alternative GPT header's LBA address. Following that, the first useable LBA sector of any partition is represented by the following eight bytes, "22 00 00 00 00 00 00 00".

The first eight bytes of the next line, "DE BF E0 01 00 00 00 00", represent the final valid LBA sector of any partition. Globally unique identifiers (UUIDs) for the physical drive are contained in the next 16 bytes, which are "37 A8 80 62 35 6B 8D 40 97 01 64 9A 87 29 0E F1".

Last but not least, the next eight bytes "02 00 00 00 00 00 00 00" represent the starting LBA of the array of GUID Partition entries, which is normally found in sector 2 for the primary partition.

In order for the system to recognize and handle disk partitions, the GPT header contains crucial details about the partition table and its structure.

Security Features, Mitigation, Vulnerabilities:

GPT, or GUID Partition Table, provides several security features that help mitigate the vulnerabilities present in MBR, or Master Boot Record. One of the most significant security features of GPT is the presence of a backup partition table. This means that if the primary partition table is corrupted or compromised, the backup partition table can be used to restore the original partition structure.

Another security feature of GPT is the CRC32 checksum, which is used to verify the integrity of the partition table. This feature helps ensure that the partition table has not been tampered with or corrupted by malicious software. Additionally, GPT can support up to 128 partitions, which provides greater flexibility in partitioning a drive and reduces the need to use multiple drives or partitions.

Furthermore, the MBR partition table is restricted to four main partitions, which might limit drive partitioning and force the usage of less secure extended partitions.

In summary, GPT security features like as backup partition tables and CRC32 checksum minimize MBR vulnerabilities while providing better protection and flexibility when partitioning a disk.

Conclusion:

Finally, GPT (GUID Partition Table) is a more sophisticated and safe partition system than MBR (Master Boot Record). Although GPT disks cannot be utilized with previous systems of operation such as Windows XP, GPT segments are supported for data storage in contemporary operating systems such as Windows 7 and above or Windows Server 2003 and above. Furthermore, only the 64-bit versions of these operating systems enable booting from UEFI-based platforms. GPT is the recommended solution for UEFI-based devices because to its improved flexibility, security, and efficiency. Managers should use GPT whenever feasible if they are mindful of the distinctions between MBR and GPT. In the end, it is apparent that GPT is the clear victor in terms of attributes and should be selected over MBR.

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